

FSK176: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2015

Chapter 4: Motion in Two Dimensions / Hfst 2: Beweging in Twee dimensies

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P4.9 (a) $a_x = 0.800 \text{ m/s}^2$; $a_y = -0.300 \text{ m/s}^2$ (b) $\theta = 339^\circ$ from $+x$ axis
(c) $x_f = 360 \text{ m}$; $y_f = -72.7 \text{ m}$; $\theta = -15.2^\circ$

P4.15 $\theta_i = 53.1^\circ$

P4.21 $h = d \tan \theta_i - \frac{gd^2}{2v_i^2 \cos^2 \theta_i}$

P4.28 (a) $\frac{v_i \sin \theta}{g}$ (b) $h + \frac{(v_i \sin \theta)^2}{2g}$

P4.29 (a) $x_i = 0.00 \text{ m}$; $y_i = 0.00 \text{ m}$ (b) $v_{xi} = 18.0 \text{ m/s}$; $v_{yi} = 0 \text{ m}$

(c) Free fall motion, with constant downward acceleration, $g = 9.80 \text{ m/s}^2$

(d) Constant velocity motion in the horizontal direction (e) $v_{xf} = v_{xi}$; $v_{yf} = -gt$ (f) $x_f = v_{xi}t$; $y_f = -\frac{1}{2}gt^2$ (g) 3.19 s (h) $v_f = 36.1 \text{ m/s}$; $\theta_f = -60.1^\circ$

P4.38 (a) 6.00 rev/s (b) $1.52 \times 10^3 \text{ m/s}^2$ (c) $1.28 \times 10^3 \text{ m/s}^2$

P4.50 (a) $2.02 \times 10^3 \text{ s}$ (b) $1.67 \times 10^3 \text{ s}$ (c) Swimming with the current does not compensate for the time lost swimming against the current.

P4.54 (a) straight up, at 0° to the vertical (b) 8.25 m/s (c) a straight up and down line (d) a symmetric parabola opening downward (e) 12.6 m/s ; 41.0° above the horizontal

P4.73 (a) $x(v_i) = (v_i \sqrt{0.1643 + 0.002299v_i^2} + 0.04794v_i^2) \text{ m}$ (b) 0.0410 m

(c) 961 m (d) $0.405 v_i$ (e) $0.0959 v_i^2$ (f) Explanation

P4.74 (a) $\theta = 26.6^\circ$ (b) 0.949